SecureRide: Detecting Safety-Threatening Behavior of E-Scooters Using Battery Information

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Reckless usage of electric (e-) scooters causes many injury accidents, raising critical safety concerns. Despite newly introduced regulations, specifically, speed limits and sidewalk driving prohibitions, the number of accidents increases due to the challenges in enforcement. Therefore, a reliable method to detect safety-threatening illegal behaviors of e-scooters is essential to mitigate this growing problem. In this article, we propose SecureRide, a system that accurately detects illegal e-scooter behaviors, i.e., speeding violation and sidewalk riding, at runtime using only battery information, without the need for additional sensors. To this end, we first design a neural network-based illegal behavior predictor that takes sequences of three battery factors, i.e., voltage, current, and capacity, as inputs. The model architecture is optimized based on time constraints, target accuracy, and resource constraints of the target devices. Next, we devise a runtime detection strategy to achieve both high accuracy and low detection time. SecureRide operates in two modes with different predictors–*lightweight-quick* and *complex-accurate* models–depending on the driving situation, ensuring both high accuracy and low detection time. We extensively validate SecureRide based on actual driving experiments. Our results show that SecureRide detects illegal behaviors with an accuracy of up to 99.77% within 1.01 seconds.

CCS Concepts: • **Computer systems organization → Embedded systems**; • **Computing methodologies → Machine learning approaches;**

Additional Key Words and Phrases: Electric scooter, Illegal behavior detection, Battery, Deep learning

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1 Introduction

Electric (e-) scooters have become a prominent mode of last-mile transportation, largely driven by the growth of sharing systems. The increasing use of e-scooters has replaced numerous short-distance car trips, significantly contributing to the reduction of carbon dioxide (CO2) emissions [12]. However, the widespread adoption of e-scooters has also raised critical safety concerns. Due to their high mobility and compact form factor, e-scooters encourage reckless riding behaviors, such as riding on pedestrian sidewalks at high speeds, which pose a serious threat to public safety. According to statistics released in the U.S., between 2017 and 2021, 169,300 safety incidents related to e-scooters were reported across the country, with the number of accidents increasing each year [27]. Consequently, many countries and cities have introduced strict regulations to prohibit safety-threatening behaviors of e-scooters, such as speeding and riding on sidewalks [11, 28].

Unfortunately, despite the introduction of regulations, the number of accidents continues to rise due to challenges in enforcement. The current enforcement of illegal e-scooter behaviors is sporadically conducted by police officers, which allows for easy evasion and proves ineffective. Due to unresolved safety concerns and the continually increasing number of incidents each year, France is even considering a complete ban on e-scooter sharing systems in Paris [4]. To fundamentally address this problem, automated enforcement is necessary, powered by an effective and practical solution for detecting illegal behaviors of e-scooters.

While specific e-scooter regulations vary across countries, speeding and riding on sidewalks are two universally prohibited behaviors. Detecting such illegal behaviors poses practical challenges in implementation. The simplest method involves using sensors. For instance, speeding can be detected with an **inertial measurement unit** (**IMU**) on e-scooters. However, e-scooter users easily bypass speed limits by manipulating the sensor values. On the other hand, Bird, one of the leading e-scooter sharing companies, has attempted to detect sidewalk riding by installing high-performance GPS and camera modules [5]. However, detection methods using additional sensors result in high installation and maintenance costs, which is not practical for sharing system companies facing profitability issues [7]. Even with the additional sensors, accurate detection remains a significant challenge. Since e-scooters are primarily used in outdoor environments, GPS and image data suffer from significant accuracy loss due to physical obstructions and weather. Therefore, there is a strong need for detection methods that overcome the inherent limitations of such sensors.

To develop a practical and effective detection method, several key challenges exist. First, practical detection should avoid the use of additional sensors and should remain robust against manipulation or environmental interference. Due to the cost-sensitive nature of e-scooter manufacturing, e-scooters typically lack sophisticated sensing capabilities and the range of available sensor data is highly limited. In addition, **microcontrollers** (**MCUs**) on e-scooters are severely resource-constrained, making it challenging to execute complex detection tasks. Next, effective detection depends on delivering timely and accurate responses during operation. E-scooters are fast-moving vehicles often used in dense urban areas where safety-critical decisions need to be made within a short response window. A detection system with low responsiveness may fail to prevent risky behavior, while one with low accuracy can generate incorrect decisions, undermining reliability.

In this article, we propose SecureRide, a battery-enabled system for detecting illegal e-scooter behaviors. SecureRide consists of two main stages: illegal behavior predictor design and runtime detection strategy. First, we propose a technique for predicting illegal behaviors using only battery information, eliminating the risk of sensor manipulation and the need for additional sensors. To design a predictor that is both accurate and lightweight, we adopt the following strategy. We first identify multiple battery factors that are highly relevant to the illegal behaviors, and feed them into the predictor as sequences to provide richer temporal context. Specifically, we analyze the e-scooter

battery states under various conditions and define three key battery factors, i.e., voltage, current, and capacity, as input features. By capturing their complementary temporal patterns as sequences, the model can better contextualize the input dynamics, leading to more accurate behavior prediction. At the same time, we simplify the output space by formulating the problem as a binary classification task, thereby reducing the decision complexity and enabling the use of compact models.

Next, predictor accuracy and detection time heavily depend on the input sequence length; longer sequences provide higher accuracy but also increase detection time, leading to an inherent tradeoff between high accuracy and fast responsiveness. To address this tradeoff, we design heterogeneous **deep neural networks** (**DNNs**), consisting of *lightweight-quick* and *complex-accurate* models, each optimized for detection time and accuracy based on different sequence lengths. Then, we propose a runtime detection strategy, called *dual-mode detection*, which leverages heterogeneous DNNs to achieve both high accuracy and low detection time. SecureRide dynamically switches between *Quick* and *Accurate* modes, utilizing lightweight-quick and complex-accurate models, respectively. We validate SecureRide with extensive actual driving experiments and on embedded platforms with limited computational resources. The key contributions of our work are as follows:

- We present the first system that detects illegal e-scooter behaviors using only battery information, without relying on any external sensors. We analyze how battery factors correlate with two major violations—speeding and sidewalk riding—and use this insight to design an accurate DNN-based predictor that operates on three battery factors: voltage, current, and capacity.
- To enable effective detection in real-world scenarios, we address the inherent tradeoff between accuracy and detection time by proposing a dual-mode detection strategy based on heterogeneous DNNs. The system dynamically switches between Quick and Accurate modes based on driving behavior, achieving both timely response and high detection precision.
- We validate SecureRide with real-world experimental data on three embedded platforms with different resource constraints, demonstrating that our system achieves an accuracy of up to 99.77% within 1.01 seconds.

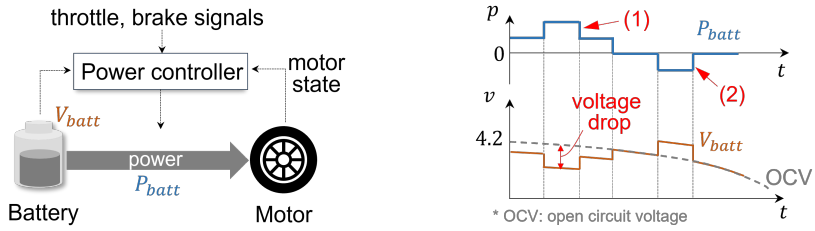
2 Background and Motivation

We describe the e-scooter illegal behaviors that threaten safety and the limitations of existing detection methods. Then, we analyze the e-scooter power system operation and battery response under these illegal behaviors.

2.1 Illegal Behaviors of Electric Scooter

Most countries and cities regulate illegal e-scooter behaviors to address growing safety concerns. The illegal behaviors commonly defined in EU countries and U.S. states fall into two: speeding and riding on pedestrian sidewalks [10, 28]. First, e-scooters must not exceed speeds of 20 or 25 km/h to prevent accidents caused by speeding. Additionally, e-scooters are required to operate only in designated lanes, including bicycle lanes, while riding on sidewalks is strictly prohibited to prevent collisions with pedestrians.

Traditionally, speeding violations and sidewalk riding are detected using sensors equipped on e-scooters. Speed is measured through an IMU, which includes accelerometer and gyroscope sensors, and speeding violations are prevented by limiting the scooter's speed. Additionally, sidewalk riding is detected by leveraging GPS data. However, these methods have inherent limitations. Firstly, speed values from IMU sensors are vulnerable to manipulation, either by hacking the firmware or physically removing the sensors. A significant number of users easily bypass the speed restrictions, allowing them to engage in illegal activities with little effort [3]. Furthermore, the accuracy of GPS signals is severely compromised by surrounding objects. Since e-scooters are often used in crowded



(a) E-scooter power system (b) Li-ion battery characteristics

Fig. 1. E-scooter power system and battery characteristics.

urban areas, the accuracy is further degraded by buildings and various road structures, such as underpasses. Overall, the critical limitations of existing methods highlight the need for a reliable and accurate detection system.

2.2 Electric Scooter Power System

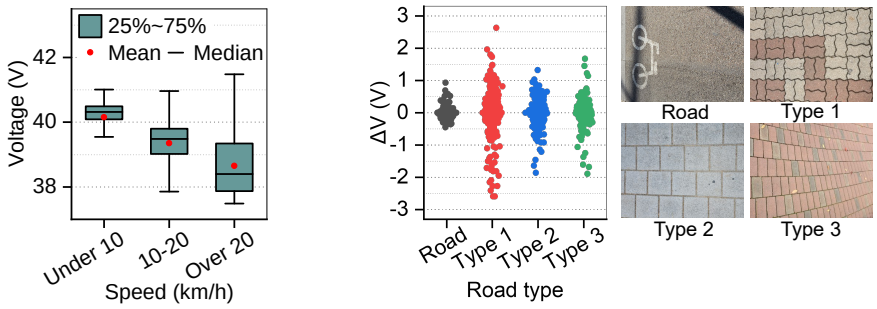
E-scooter power systems consist of a battery, an electric motor, and a power controller. They operate by delivering power from the battery to the motor, as shown in Figure 1(a). The power controller manages the power flow based on user inputs, such as throttle and brake signals, as well as the battery and motor states.

Impact of e-scooter speed on battery voltage. Lithium-ion (Li-ion) batteries are widely used in e-scooter applications due to their high energy density and long lifespan. Figure 1(b) illustrates the voltage characteristics of Li-ion batteries. The battery terminal voltage drops when the battery supplies power and increases when it receives power from external sources. This indicates that the battery voltage fluctuates based on the throttle control by the user. When the user adjusts the throttle, the power consumed by the electric motor changes, creating various power patterns during driving. For instance, when the throttle is engaged to increase speed, the power consumption from the battery rises sharply, as indicated by point (1) in the figure, resulting in a voltage drop. Conversely, when braking or decelerating, regenerative braking is activated, causing the electric motor to operate in reverse. This results in current flowing back to the battery, as indicated by point (2), raising the voltage. These phenomena suggest that the battery voltage reflects speed changes driven by the user's throttle and brake inputs.

Impact of road surface on battery voltage. When an e-scooter travels over pedestrian sidewalks paved with blocks, the physical irregularities introduce repetitive vibrations. Unlike smooth asphalt roads, these surfaces cause small-scale but frequent disturbances in the scooter's movement, especially due to the small wheel size (approximately 8 inches) typical of e-scooters. These disturbances lead to instantaneous changes in the motor's power demand, as the control system continuously compensates for rolling resistance and uneven ground contact. As a result, the power drawn from the battery fluctuates irregularly, and as shown in Figure 1(b), this power fluctuation directly influences the battery voltage—an effect also observed under vibration conditions in prior studies [13, 29]. This indicates that the battery voltage reflects the road conditions, such as rough or uneven surfaces, and the type of terrain the e-scooter is operating on. For instance, smoother asphalt roads produce fewer voltage fluctuations compared to cobblestone or bumpy surfaces.

2.3 Battery Voltage Response under Illegal Behaviors

In Section 2.1, we described two typical illegal behaviors of e-scooters, i.e., speeding violation and sidewalk riding. We then analyzed the impact of e-scooter speed and road surface on the battery



(a) Speeding violation (b) Sidewalk riding

Fig. 2. E-scooter illegal behaviors and battery voltage response. Battery voltage exhibits different patterns depending on speed and road surfaces.

voltage in Section 2.2. Based on this analysis, we conducted real-world e-scooter driving experiments to observe the battery voltage under the illegal behaviors. We conducted the driving experiments using a Xiaomi M365, a widely used commercial e-scooter. The details of the experimental setup are described in Section 6.1.

Speeding violation. To observe the impact of e-scooter speed on battery voltage, we measured the voltage while driving at various speeds. Figure 2(a) illustrates the voltage distribution corresponding to different speed ranges: under 10, 10–20, and over 20 km/h. As shown in the figure, the battery voltage decreases as the e-scooter speed increases. Notably, we can observe a significant difference between the speed ranges; when speeding over 20 km/h, the voltage dropped by approximately 9.5% compared to speeds under 20 km/h. This result can be explained by the fact that higher speeds require more power to maintain motion, as the scooter needs to overcome greater air resistance and friction. The increased power leads to a more noticeable drop in battery voltage, as indicated by (1) in Figure 1(b). Therefore, battery voltage naturally reflects the scooter's speed through its response to changing power demands.

Sidewalk riding. We observe the impact of different road surfaces on battery voltage. We measured the voltage while driving on various types of roads, including pedestrian sidewalks. Figure 2(b) shows a distribution of voltage difference ($\Delta V = V(t+1) - V(t)$) on four types of roads; "Road" refers to a bicycle road, and Type 1, Type 2, and Type 3 represent different pedestrian sidewalks. As seen in the graph, unlike the bicycle road, the variance of ΔV is larger on pedestrian sidewalks. While the driving road is typically paved uniformly with asphalt, sidewalks are constructed with paving blocks that introduce frequent surface irregularities. These irregularities cause small but rapid changes in the motor's power demand, leading to fluctuations in battery voltage. The data further shows that the variance of ΔV differs across sidewalk types. Based on our observation, the smaller and more segmented the blocks are, the greater the variance in ΔV. These results demonstrate that battery voltage not only reflects coarse terrain differences (e.g., road vs. sidewalk), but also captures fine-grained variations in sidewalk construction and surface pattern.

2.4 Summary and Challenges

We addressed two key issues related to illegal e-scooter behaviors. First, we identified two commonly regulated illegal behaviors globally—speeding and sidewalk riding—and discussed the limitations of existing sensor-based detection methods. Speed values from IMU sensors can be easily hacked, and the accuracy of GPS data for sidewalk detection is significantly degraded depending on surrounding

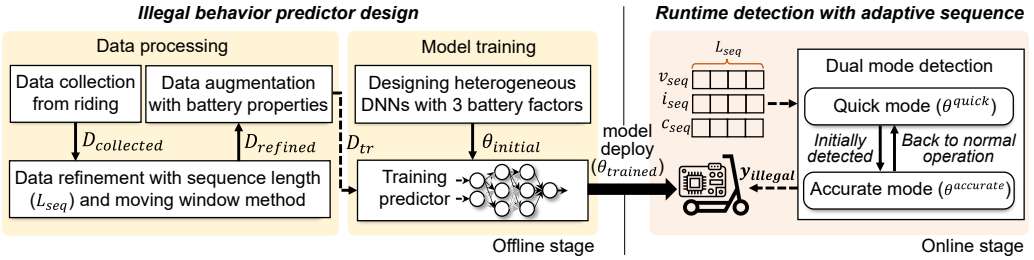


Fig. 3. SecureRide overview.

objects. Second, we explored the feasibility of detecting illegal behaviors using battery information through real-world driving experiments. The battery voltage changes depending on the speed and physical vibrations of e-scooters, which are affected by speeding violations and sidewalk riding, respectively.

We have two technical challenges in detecting illegal e-scooter behaviors using battery information. First, developing an accurate predictor that is resistant to malicious manipulation is a challenge. Li-ion battery operations rely on electrochemical reactions, which are influenced by factors such as temperature and **state of charge** (**SoC**). Therefore, creating an accurate predictor that accounts for these complex physical characteristics is highly challenging. Additionally, a robust method to prevent manipulation is required. Second, achieving high responsiveness in runtime detection is also challenging. Since e-scooters use MCUs with limited computational and memory resources, algorithms with heavy computational complexity are impractical. Furthermore, the detection time should be low to enable real-time enforcement. SecureRide addresses both of these technical challenges.

3 SecureRide Overview

We propose SecureRide, a system that accurately detects illegal e-scooter behaviors at runtime. Figure 3 outlines the overview of SecureRide. SecureRide consists of offline and online stages. In the offline stage, SecureRide provides a predictor ($\theta_{trained}$) that directly detects the presence of illegal behavior, based on sequences of three battery factors, i.e., voltage, current, and capacity. In the online stage, SecureRide operates the predictor with adaptive sequence length in two modes to achieve both high accuracy and low detection time.

The first stage, the illegal behavior predictor design, encompasses the entire offline process, including data processing and model training. During the data processing phase, in the data collection, battery information, speed, and GPS data are selectively collected during e-scooter driving to ensure high-quality data $D_{collected}$. In the data refinement module, battery factors are extracted from $D_{collected}$ and split into 1D vectors with various sequence lengths. The data is then labeled using the speed and GPS data. The refined data ($D_{refined}$) is provided to the data augmentation module, where data augmentation based on the physical properties of the battery is applied, resulting in the final dataset (D_{tr}). Meanwhile, in the model training phase, heterogeneous DNNs ($\theta_{initial}$), consisting of a lightweight-quick model and a complex-accurate model, are designed, each optimized for time constraints and accuracy. The input (v_{seq}, i_{seq}, c_{seq}) to the DNNs is sequences of three battery factors—voltage, current, and capacity—and the output ($y_{illegal}$) is binary values, i.e., true or false, indicating the presence of illegal behaviors. The designed predictor $\theta_{initial}$ is trained using the dataset D_{tr}, finally, resulting in the predictor $\theta_{trained} = \{\theta^{quick}, \theta^{accurate}\}$.

The second stage includes a runtime detection strategy, namely, dual-mode detection, which ensures both high accuracy and low detection time. The designed heterogeneous DNNs, θ^{quick}

and $\theta^{accurate}$, use different input sequence lengths (L_{seq}^{short} and L_{seq}^{long}), which significantly impact both detection accuracy and time. Our experiments show that longer input sequences lead to higher accuracy (see Figure 5) but also result in longer detection times. Longer detection times cause increased missed detection points, posing a key drawback for capturing fleeting moments in e-scooter riding. If we use a sequence obtained over 1 s as the input, detection can be achieved in around 1 s, whereas using a 5-s sequence would result in a 5-s detection time. To overcome the tradeoff, we propose a dual-mode detection technique. SecureRide operates in two different modes: Quick mode, which uses θ^{quick} with L_{seq}^{short}, and Accurate mode, which uses $\theta^{accurate}$ with L_{seq}^{long}. During driving, the system defaults to Quick mode with a shorter sequence (L_{seq}^{short}) to rapidly detect illegal behaviors. Once an illegal behavior is detected, the system switches to Accurate mode with $\theta^{accurate}$. If Accurate mode also detects the illegal behavior, it is confirmed. If the behavior is not detected in Accurate mode, the previous detection in Quick mode is ignored and the system reverts to Quick mode. This way, we ensure both accuracy and responsiveness in detection.

Advantages of SecureRide. SecureRide offers several advantages compared to traditional detection methods. First, SecureRide significantly reduces the risk of sensor manipulation by relying solely on battery telemetry, rather than externally exposed sensors such as IMU. Battery parameters—such as voltage, current, and capacity—are monitored through embedded **power management ICs (PMICs)** or BMS chips that are integrated within the battery pack and not directly accessible via external interfaces. These signals are essential for real-time power regulation (see Figure 1(a)), and incorrect values may disrupt motor operation or trigger ride-stopping protections (e.g., overcurrent limiters and undervoltage cutoff). While it is theoretically possible for a firmware-level attacker to alter battery readings for inference-only manipulation, such an attack requires privileged access and careful control to avoid interfering with power delivery logic. This complexity and risk make inference-time manipulation significantly more difficult compared to spoofing IMU sensors. Therefore, although not entirely immune to adversarial inputs, SecureRide leverages the tightly coupled nature of battery sensing and control to enhance trustworthiness in behavior detection.

Second, SecureRide also addresses the limitations of GPS-based methods, which suffer from degraded accuracy due to signal obstruction in urban environments or poor weather conditions. In contrast, battery factors are collected from embedded PMICs and are unaffected by external geometry or signal interference, ensuring more consistent and reliable detection performance.

Usage Scenarios of SecureRide. SecureRide operates in two distinct stages—offline and online—each serving different roles and stakeholders. In the offline stage, the primary goal is to develop a robust illegal behavior predictor by training deep learning models using battery data. During this phase, speed and GPS data are additionally used solely for labeling purposes, enabling accurate supervision in the training process. We expect that this stage is typically performed by operators or manufacturers who have access to full sensor logs. In contrast, the online stage targets real-time operation in deployed e-scooters. The trained predictor operates solely on battery information—voltage, current, and capacity—collected from PMICs on the scooter to detect the illegal behaviors.

4 Illegal Behavior Predictor Design

We describe the data processing phase, including collection, refinement, and augmentation, followed by the model training phase, which covers model design and training.

4.1 Data Processing

4.1.1 Data Collection. The goal of our data collection module is to ensure high data quality. SecureRide requires battery factors, speed values, and GPS data to build a dataset. Battery factors such as voltage, current, and remaining capacity can be measured from battery monitor chips

typically integrated into the PMICs. We obtain the speed values from IMU sensors. Unlike battery factors and speed, GPS data accuracy degrades with the driving environment, which can result in incorrect labeling of sidewalk riding detection. For example, GPS experiences a significant increase in the **estimated position error** (**EPE**) when **line of sight** (**LOS**) is not available. To address this issue, we impose restrictions during data collection. First, data is only collected when the EPE value falls below a threshold ϵ. For example, if ϵ is 3, only data with an error of less than 3 meters is collected. The following equation describes the restriction:

$$D_{collected} = \{d(k) \in D_{raw} | EPE(k) \leq \epsilon\}. \tag{1}$$

D_{raw} is raw data, k is its index, and $D_{collected}$ is the collected data. Additionally, to clearly distinguish between driving roads and sidewalks in each data point, we limit the data to cases where the area within the EPE radius around the estimated position falls within the same road type (i.e., sidewalk or road). The EPE value is part of the GPS data. This restriction prevents incorrect labeling caused by misclassifying road types due to GPS inaccuracy, ensuring the quality of the data.

4.1.2 Data Refinement. The goal of the data refinement module is to convert the collected data into input-output formats that are effective for prediction. We first refine the input data to effectively represent the battery state and enhance data diversity. Since the battery state is not temporally independent, it can be accurately represented in a sequence format. For example, battery voltage varies depending on the SoC, and each data point is influenced by previous values. Therefore, for the battery factors, we process the data into 1-D sequences of a specific length L_{seq}. We enhance diversity by slicing the data with a moving window and a minimum step size (default 1). Our data refinement for battery factors is

$$\begin{aligned} D_{refined} = \{D_{collected}[k : k + L_{seq}]\} \\ \text{s.t. } 0 \leq k \leq |D_{collected}| - L_{seq} \end{aligned}. \tag{2}$$

The output of the predictor is a binary value, either true or false, indicating the presence of illegal behavior for each sequence. A "true" means the e-scooter is in an illegal behavior state, while "false" indicates the opposite. We assign "true" when the e-scooter exhibits illegal behavior at all points within the input sequence data. For example, for an input sequence of battery factors with length L_{seq}, if the speed at all time points exceeds the threshold, the output is "true." Otherwise, the output is "false." We structure the input in sequence format to effectively learn complex battery characteristics, while simplifying the output to binary labels to enhance prediction efficiency.

4.1.3 Data Augmentation. The goal of our data augmentation is to improve the generalizability of the predictor by considering battery properties during augmentation. As described in Section 2.2, several factors such as throttle control, braking, and road type affect power consumption and, consequently, battery voltage. Based on this analysis, we implement data augmentation injecting noise that reflects these characteristics. We first apply Gaussian noise, which simulates naturally occurring noise. Then, we use spike noise to inject short, sharp changes at specific points, which accounts for throttle adjustments by the user or sudden power changes due to road conditions (e.g., bumps and manholes). In addition, step and slope noise are applied to create gradual or linear changes in the sequence, covering scenarios with varying road inclines. This way, we perform appropriate data augmentation reflecting battery properties under various driving environments, as expressed by the following equation:

$$D_{tr} = F_{GN}(D_{refined}) \cup F_{SN}(D_{refined}) \cup F_{SS}(D_{refined}). \tag{3}$$

F_{GN} is the function injecting Gaussian noise, F_{SN} is for spike noise, and F_{SS} is for step and slope noise.

4.2 Model Training

We design heterogeneous DNNs with the three battery factors and train them with the dataset D_{tr}.

4.2.1 Designing Heterogeneous DNNs with Three Battery Factor. We have two key ideas for designing the illegal behavior predictor: first, we use the three battery factors as inputs for effective prediction; second, we design heterogeneous DNNs composed of two models with different performance levels, considering the resource constraints of the target devices.

Three battery factors. We described the changes in battery voltage depending on power consumption in Section 2.2 with Figure 1(b). The following equation shows the battery voltage:

$$V_{batt}(t) = V_{OCV}\{SoC(t)\} - I_{batt}(t) \cdot R_{internal}(t). \tag{4}$$

V_{batt} indicates battery voltage, SoC state of charge, V_{OCV} open circuit voltage, I_{batt} current draw, and $R_{internal}$ internal resistance of the battery. As shown in the equation, the battery voltage is determined by subtracting the product of the current drawn and the battery's internal resistance from the open circuit voltage, which depends on the remaining capacity. This means that the battery voltage is directly affected by the current and the remaining capacity and these three factors are interrelated, reflecting the overall battery state. Therefore, SecureRide uses these three factors to accurately capture the battery state. Voltage and current values can be obtained from the PMIC installed in any commercial e-scooter. The remaining capacity is calculated using the Coulomb counting method, which is also typically available from PMICs.

Designing Heterogeneous DNNs. We design an illegal behavior predictor using sequences of the three battery factors as inputs. Li-ion batteries exhibit complex and nonlinear voltage characteristics due to their operation via electrochemical reactions. Additionally, as mentioned in Section 4.1.2, the battery state is temporally dependent, meaning that the battery factors are influenced by values from adjacent time points. To address the complex nature of battery properties and temporal dependencies (inductive bias), we employ a DNN technique, specifically 1D **convolutional neural networks** (**CNNs**), which are capable of learning the temporal patterns between adjacent data points to build our predictor. Model selection is discussed in detail in Section 6.2.1.

Accuracy and detection time are key performance metrics for illegal behavior detection. The model design principle of SecureRide is to develop heterogeneous DNNs consisting of two models with different performance levels, considering the resource constraints of the target devices. The lightweight-quick model (θ^{quick}) is optimized for time constraints while the complex-accurate model ($\theta^{accurate}$) is optimized for accuracy. The quick model has low computational complexity, consumes fewer resources, and provides high responsiveness. Since the feature of this model is high responsiveness, the model architecture (θ^{quick}) and its input sequence length (L_{seq}^{short}) are determined to ensure compliance with the predefined detection time constraint ($\tau_{constraint}$):

$$T_{execution}(\theta^{quick}) + T_{buffer}(L_{seq}^{short}) \leq \tau_{constraint}. \tag{5}$$

$T_{execution}(\theta^{quick})$ refers to the time required for model inference, and $T_{buffer}(L_{seq}^{short})$ is the time needed to receive and buffer a sequence of battery factors of length L_{seq}^{short} from the PMIC. We first determine θ^{quick} that should have a small number of hidden layers to ensure low computational complexity. In our work, we use one hidden layer for θ^{quick}. Then, we maximize L_{seq}^{short} while ensuring it satisfies the time constraint.

Meanwhile, the accurate model should have high accuracy. We determine the architecture of $\theta^{accurate}$ and the input sequence length L_{seq}^{long} based on the target accuracy as follows:

$$Accuracy(\theta^{accurate}, L_{seq}^{long}) \geq \alpha_{target}. \tag{6}$$

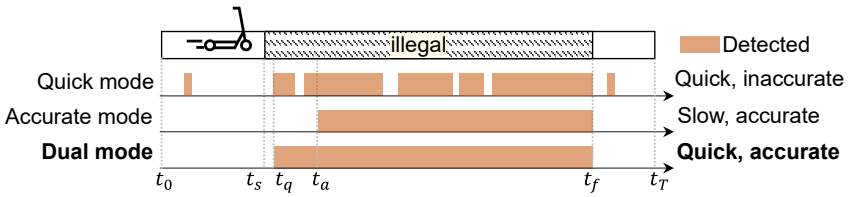


Fig. 4. Dual-mode detection.

MCUs in e-scooters have highly limited computation and memory resources. Therefore, we maximize the number of hidden layers in $\theta^{accurate}$ while ensuring the target device can handle the model's execution. Then, we minimize L_{seq}^{long} while still achieving the target accuracy.

4.2.2 Training Predictor. The designed heterogeneous DNNs (θ^{quick} and $\theta^{accurate}$) are trained using dataset D_{tr}. Each model is trained using refined data $x_i^{short}, x_i^{long} \in D_{tr}$, which correspond to their respective input sequence lengths, L_{seq}^{short} and L_{seq}^{long}, as follows:

$$\theta^{quick} = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(f(x_i^{short}; \theta^{quick}), y_i), \tag{7}$$

$$\theta^{accurate} = \arg\min_{\theta} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}(f(x_i^{long}; \theta^{accurate}), y_i). \tag{8}$$

N is the number of data samples, $f(x_i); \theta$ is the prediction, and y_i is the label.

5 Runtime Detection with Adaptive Sequence

SecureRide detects illegal e-scooter behaviors with high accuracy and low detection time. To achieve this, we propose a dual-mode detection strategy. We describe the tradeoff between accuracy and detection time based on heterogeneous DNNs, then introduce the techniques we developed to overcome this tradeoff.

5.1 Accuracy and Detection Time Tradeoff

The heterogeneous DNNs include θ^{quick} and $\theta^{accurate}$. θ^{quick} has a shorter detection time but lower accuracy, whereas $\theta^{accurate}$ has higher accuracy but also a longer detection time. In SecureRide, we refer to using θ^{quick} as Quick mode and using $\theta^{accurate}$ as Accurate mode. Figure 4 shows the operation of these two modes. We assume that an e-scooter travels over the period from t_0 to t_T, with illegal behavior occurring during the hatched area (from t_s to t_f). With Quick mode (Quick), which uses short sequences, we can detect the illegal behavior from t_q due to its low detection time. However, with lower accuracy, detection is unreliable in the illegal region, and false positives occur in the non-illegal region. Meantime, Accurate mode with longer sequences begins detecting the illegal behavior from t_a due to its high detection time. Instead, illegal behaviors are accurately detected throughout the entire period. This shows the tradeoff between accuracy and detection time when using these two models.

5.2 Dual-mode Detection

We propose a dual-mode detection technique to achieve high accuracy and low detection time, overcoming the tradeoff. The key idea is to switch between modes based on detection events to compensate for their limitations. In dual-mode detection, Quick mode is the default, but once

ALGORITHM 1: Dual-mode Detection

Require: Heterogeneous DNNs θ^{quick} and $\theta^{accurate}$

1: /* Main */
2: $\theta \leftarrow \theta^{quick}$; $L_{seq} \leftarrow L_{seq}^{short}$
3: $y \leftarrow$ `False`
4: **while** during riding
5: $v_{seq}, i_{seq}, c_{seq} \leftarrow getInput(L_{seq})$
6: $y_{illegal} \leftarrow \theta(v_{seq}, i_{seq}, c_{seq})$
7: **if** $y_{illegal} =$ `True` **and** *Quick* mode **then**
8: $\theta \leftarrow \theta^{accurate}$; $L_{seq} \leftarrow L_{seq}^{long}$ ▷ Switch to Accurate mode
9: **else if** $y_{illegal} =$ `False` **and** *Accurate* mode **then**
10: $\theta \leftarrow \theta^{quick}$; $L_{seq} \leftarrow L_{seq}^{short}$ ▷ Switch to Quick mode
11: $y \leftarrow y_{illegal}$; action(y)
12: /* Getting runtime input with a sequence length */
13: **function** *getInput*(L)
14: $v, i, c \leftarrow$ get sensor values each with a length of L
15: **return** v, i, c

illegal behavior is detected, the system switches to Accurate mode for better accuracy. Algorithm 1 details the proposed technique. First, SecureRide sets Quick mode with the quick model (θ^{quick}) as the default (line 2). This model has low computational complexity, making it highly efficient, and the short sequence length required for detection allows for high responsiveness. Once driving begins, SecureRide operates in Quick mode, receiving sequences of three battery factors with L_{seq}^{short} from PMIC and detecting illegal behavior (lines 5 and 6). If illegal behavior is detected (line 7), the system switches to Accurate mode using the accurate model ($\theta^{accurate}$) (line 8). For the first detection in Accurate mode, the system uses an input sequence of L_{seq}^{long} combining the last Quick mode input with additional sensed values. If Accurate mode returns a false result (line 9), we assume the initial Quick mode detection was an error or the illegal behavior has ended. Therefore, the system returns to Quick mode (line 10). Each prediction result, y, is provided for further actions, such as logging or raising an alert, based on the policy when illegal behavior is detected (line 11).

The proposed dual-mode detection offers several key advantages. First, illegal behaviors are detected with both high accuracy and low detection time. Quick mode provides low detection time but low accuracy, leading to unreliable detection. Accurate mode offers consistent detection, but its high detection time reduces responsiveness, a key drawback during fleeting moments in e-scooter riding. Dual-mode detection ensures high accuracy and responsiveness by leveraging both modes. As shown in Figure 4, our approach detects illegal behavior from t_q with low detection time and maintains high accuracy. Furthermore, dual-mode detection operates efficiently on resource-constrained devices by defaulting to Quick mode. We typically use θ^{quick} for low computational complexity, activating $\theta^{accurate}$ only when needed.

6 Evaluation

We validate SecureRide by collecting real-world data, training a predictor, and deploying it on embedded platforms.

Table 1. Data Distribution across Road Types (Types 1–13: Sidewalks, Road: Bicycle Roads)

Road	**%**	**Road**	**%**	**Road**	**%**	**Road**	**%**	**Road**	**%**
Type 1	3.662	Type 2	3.595	Type 3	5.645	Type 4	5.594	Type 5	5.558
Type 6	5.023	Type 7	4.800	Type 8	2.287	Type 9	2.187	Type 10	2.229
Type 11	2.156	Type 12	2.096	Type 13	2.077	Road	53.092	**Total**	**100.000**

Table 2. Data Distribution across Speed Ranges

Speed Range (km/h)	0–5	5–10	10–15	15–20	20–25	**Total**
Percentage (%)	3.10	15.98	23.66	20.60	36.66	**100.00**

6.1 Preliminaries

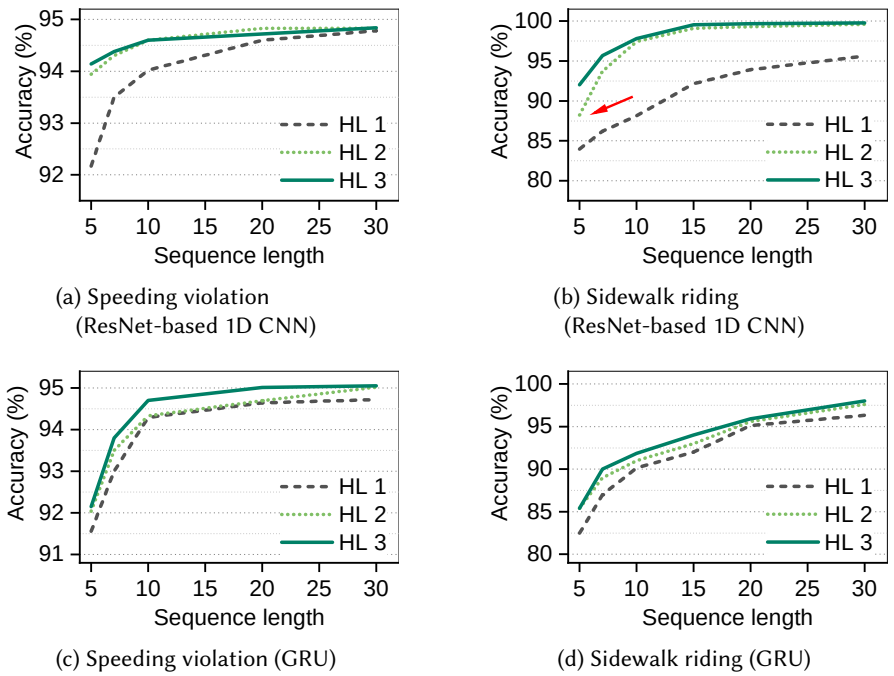
Datasets from real-world driving experiments. We collected real-world e-scooter driving data. To ensure environmental diversity, we selected 20 optimal locations. The curated locations comprise (1) designated lanes with diverse slope, including flat, uphill, and downhill sections, and (2) sidewalks with 13 different paving block patterns. To capture a wide range of user driving behaviors, we performed multiple trials at each location, varying speed, acceleration/deceleration, and SoC of e-scooter. We gathered 36 hours of driving data with the Xiaomi M365 e-scooter, commonly used in sharing systems. Tables 1 and 2 show the percentage-based data distribution. Table 1 presents the distribution across road types, where "Type 1", "Type 2", ... refer to 13 different sidewalk patterns, and "Road" denotes the bike roads. Table 2 presents the distribution across speed ranges. We further discuss the impact of data distribution in Section 8.1.

During driving, data were transmitted via the e-scooter's BLE module to a smartphone app, including battery factors, speed, and GPS. The speeding violation was identified based on e-scooter's IMU, while sidewalk riding was determined using GPS data mapped to the OpenStreetMap. In addition, we manually recorded road types during the driving experiments to obtain the ground-truth labels for sidewalk riding. We split the collected dataset into training (64%), validation (16%), and test (20%) sets using group-based splitting to avoid data overlap, while stratified sampling preserved label distribution in the validation set.

Implementation of heterogeneous DNNs. To implement the proposed predictors, we employ a lightweight architecture considering the deployment constraints of embedded devices such as e-scooters, specifically a ResNet-based 1D CNN. The input first passes through an attention block to highlight important features, followed by a series of residual blocks, the number of which is determined by the hidden layer setting (up to three). Each block consists of three 1D convolutional layers: the first two are followed by batch normalization, ReLU activation, and dropout, and the third connects to a shortcut path via batch normalization. A global average pooling layer and a softmax output layer are used for the final classification.

In addition, we implemented a **Gated Recurrent Unit** (**GRU**) model to evaluate the suitability of RNN-based predictors, given GRU's effectiveness in handling sequential data and its lower computational overhead compared to LSTM. The GRU model consists of stacked GRU layers based on the hidden layer setting (up to three), followed by a fully connected output layer. The models employ a softmax output layer for classification.

For training all models, we applied dropout (0.3), L2 regularization (0.01), and early stopping with a patience of 10 to enhance generalization. All models were trained using the Adam optimizer,



(a) Speeding violation (ResNet-based 1D CNN)

(b) Sidewalk riding (ResNet-based 1D CNN)

(c) Speeding violation (GRU)

(d) Sidewalk riding (GRU)

Fig. 5. Accuracy validation. SecureRide achieves high accuracy in detecting illegal behaviors using both CNN and RNN models.

categorical cross-entropy loss, and a dynamic learning rate. All implementations were based on TensorFlow 2.13.0.

Experimental platforms. SecureRide targets e-scooter applications. We validate the model using three devices, considering the performance of both the e-scooter's onboard MCUs and connected smartphone platforms: Arduino Nano 33 BLE [2], Raspberry Pi Zero W [22], and NVIDIA Jetson Orin Nano [21]. The models were converted to TensorFlow Lite (TFLite) models.

6.2 Validation of Illegal Behavior Predictor

We evaluate the accuracy and detection time for speeding violations and sidewalk riding across different model architectures. We then verify the effectiveness of each component using the sidewalk riding prediction model.

6.2.1 Accuracy. We first validate the predictor's accuracy. Figure 5(a) and (b) shows the accuracy of the ResNet-based 1D CNN models for speeding violations and sidewalk riding, respectively, with varying sequence lengths and hidden layers. We denote hidden layer as HL and sequence length as SL (e.g., HL 3 means three hidden layers are used and SL 5 means input sequence length is five). Note that the model did not have coverage for sequence lengths below five. For speeding violations, accuracy is 92.17% with the shortest sequence length (SL 5) and the minimum hidden layers (HL 1). As the sequence length and the number of hidden layers increase, the accuracy improves, reaching up to 94.84%. Meanwhile, the prediction accuracy of sidewalk riding shows a range from a minimum of 83.95% to a maximum of 99.77%. These results show that our model detects illegal behavior with high accuracy. We also observe that accuracy improves as the sequence length increases and the number of hidden layers grows. In particular, sidewalk riding accuracy improves more

Table 3. Normalized Confusion Matrices with Different Sequence Lengths (Sidewalk riding, HL2)

Actual	Predicted (HL2, SL5)		Predicted (HL2, SL10)		Predicted (HL2, SL20)	
	T	F	T	F	T	F
T	0.4315	0.0671	0.4887	0.0075	0.5058	0.0031
F	0.0659	0.4355	**0.0182**	0.4856	**0.0074**	0.4837

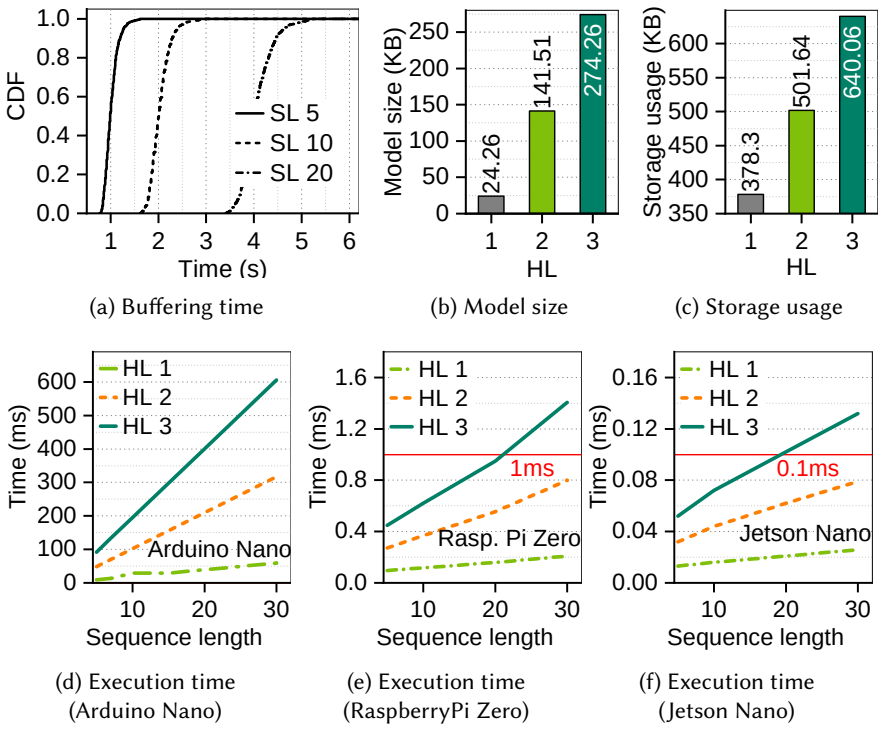
significantly than speeding violations as sequence length increases. Similarly, as indicated by the arrow, increasing the number of hidden layers boosts accuracy when using shorter sequences for sidewalk riding. These findings suggest that sidewalk riding patterns are more complex than speeding violations, making them more sensitive to sequence length and model complexity.

Next, Figure 5(c) and (d) presents the accuracy of GRU-based models for the same tasks. Consistent trends are observed, where accuracy increases with longer sequence lengths and more hidden layers, showing the applicability of our method to RNN models. However, GRU-based models exhibit lower accuracy in our experiments and offer limited compatibility with embedded ML frameworks, which are typically optimized for executing CNNs through specialized kernels. Given these advantages, we opted for CNN-based models, which are more efficient and suitable for deployment in resource-constrained environments.

For further analysis, we examine the normalized confusion matrices, where each cell represents the proportion of predictions relative to the total, for sidewalk riding detection using the 1D CNN model with HL 2 (Table 3). As sequence length increases, both false positives (Actual: F, Predicted: T) and false negatives (Actual: T, Predicted: F) decrease, indicating improved reliability. Notably, even in high-accuracy settings (SL 10 and SL 20), the false positive rate remains higher than the false negative rate, suggesting the model is conservatively biased toward over-detection. These side effects may be mitigated by applying temporal smoothing during post-verification, which helps suppress transient misclassifications and improve detection stability.

6.2.2 Detection Time. We analyze the detection time to verify the practical usability of our predictor. The detection time includes buffering time and execution time. Buffering time refers to the time taken to retrieve the battery factors from the sensors, which largely depends on the sampling rate of the e-scooter's PMICs (e.g., battery monitor). We measure the buffering time with the e-scooter used in our experiments. Figure 6(a) shows the **cumulative distribution function (CDF)** of the buffering time. As shown in the figure, the buffering time takes up to 1.5 s when SL 5, increasing linearly with longer sequences. This indicates that the sequence length should be kept as short as possible to enable fast detection. Note that although we use a commercial entry-level e-scooter, typical ICs [14, 15, 25] feature an ADC conversion time of 125 ms or less, meaning the buffering time can be reduced with better hardware.

We assess model size and execution time by deploying the models on three embedded devices: Arduino Nano 33 BLE (Arduino Nano), Raspberry Pi Zero, and Jetson Orin Nano (Jetson Nano). Figure 6(b) shows model sizes, while memory usage and execution times are shown in Figure 6(c) and (d)–(f). We observe that the model sizes varied significantly depending on the number of hidden layers, not the sequence length. Note that this demonstrates the appropriateness of our DNN design principle, which limits the number of hidden layers based on device memory constraints. Unlike the buffering time, which solely depends on the sequence length, the execution time is heavily influenced by both the sequence length and the number of hidden layers. As shown in Figure 6(d), the execution time of HL 1 on Arduino remains mostly under 50 ms regardless of sequence length, with the shortest sequence completing in 9.3 ms. However, HL 2 and HL 3 showed an execution time range of 48.6–317 ms and 92–605 ms, respectively. Figure 6(e) and (f) shows similar trends, but the execution times are extremely low—1 ms and 0.1 ms, respectively, for up to two hidden



(a) Buffering time (b) Model size (c) Storage usage

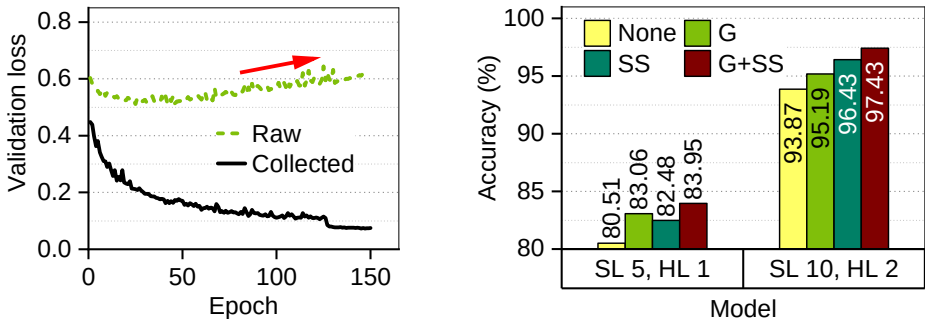
(d) Execution time (Arduino Nano) (e) Execution time (RaspberryPi Zero) (f) Execution time (Jetson Nano)

Fig. 6. Detection time. SecureRide shows acceptable detection times on devices with limited computation and memory resources.

layers. This indicates that (1) an appropriate model architecture should be selected based on the device and (2) our model can sufficiently operate in real-time with the commercial e-scooters.

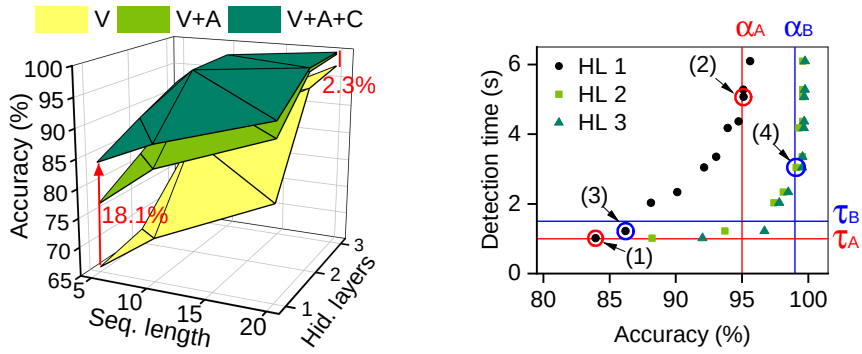
6.2.3 Effectiveness of Data Processing. SecureRide's data processing includes data collection, refinement, and augmentation. To examine the effectiveness of our data collection method, we compare the model's performance when trained on raw data (D_{raw}) versus data collected using our method ($D_{collected}$), which excludes incorrect GPS data. Figure 7(a) shows the validation loss during the model training with the same structure (SL 20 and HL 2). The validation loss diverges when using D_{raw} (Raw), while it converges stably when using $D_{collected}$ (Collected). This highlights the impact of inaccurate labels present in the raw data.

To verify the effect of our data augmentation method, we compare the accuracy of four models with the same structure but trained with different augmentations, as shown in Figure 7(b). The first model has no augmentation (None), the second uses Gaussian (G), the third applies spike, step, and slope (SS), and the last uses all (G+SS). As shown in the figure, there is a noticeable accuracy improvement with the augmentation methods, regardless of the model structure. For example, G+SS results in about 3.5% accuracy improvement. G augmentation, which adds white noise, is more effective for short sequences (SL 5, HL 1) whereas SS augmentation is more effective for longer sequences (SL 10, HL 2). This suggests that longer sequences contain more battery state information, making SS augmentation, which better reflects the physical properties of the battery, more effective. Note that the effect of data refinement has already been validated through the fact that the accuracy improves as the sequence length increases, so no additional verification is performed. Overall, the data processing phase improves the stability and accuracy of the predictor.



(a) Effect of data collection (b) Effect of data augmentation

Fig. 7. Effectiveness of data processing. SecureRide's data collection and data augmentation contribute to stable model training and high accuracy.



(a) Effect of three battery factors (b) Operation of heterogeneous DNNs

Fig. 8. Performance of heterogeneous DNNs.

6.2.4 Validation of Heterogeneous DNNs. We validate the performance of the proposed heterogeneous DNNs. We first analyze the impact of the three battery factors on model performance. We compare three cases with different usage of the factors: using only the voltage factor (V), using both voltage and current factors (V+A), and using voltage, current, and capacity factors (V+A+C). Figure 8(a) shows the accuracy of the three cases. V+A shows higher accuracy compared to V, and V+A+C shows even higher accuracy compared to V+A, demonstrating the effectiveness of our three battery factors. However, the accuracy improvement varies significantly depending on the model structure. For SL 5 and HL 1, accuracy increases from 65.85% with V, to 76.87% with V+A, and 83.95% with V+A+C, leading to an overall enhancement of 18.1%. Meanwhile, under SL 20 and HL 3, accuracy improves from 97.34% with V to 99.68% with V+A+C. This trend indicates that using more factors is particularly effective for shorter sequences, where it is more difficult to reflect the battery's state.

We proposed heterogeneous DNNs to meet the time constraint and target accuracy. To analyze the effectiveness of the method, we compare the performance of designed models, i.e., quick model (θ^{quick}) and accurate model ($\theta^{accurate}$), with models with various sequence lengths and numbers of hidden layers. We use two model cases, i.e., A and B, designed to meet different conditions. In case A, the time constraint (τ_A) and target accuracy (α_A) are set to 1 s and 95%, respectively, while in case B, they are set to 1.5 s (τ_B) and 99% (α_B). For case A, we assumed stricter memory constraints,

limiting the number of hidden layers to one, while case B is limited to two hidden layers. Our method designs the quick model with SL 5, HL 1, and the accurate model with SL 25, HL 1 for Case A, while the quick model has SL 7, HL 1, and the accurate model SL 15, HL 2 for Case B. We perform 1,000 predictions using models with various structures, and Figure 8(b) shows the accuracy and average detection time for each model. In the figure, marks (1) and (2) represent the quick and accurate models of case A, while marks (3) and (4) represent those of case B. Notably, the quick models in each case are the most lightweight and offer the best accuracy among the models that meet the time constraints. For example, models with HL 1 that show detection times less than τ_B are marks (1) and (3). Among these options, model (3), with higher accuracy, is selected. Meanwhile, the accurate models are selected based on the shortest detection time while meeting the hidden layer constraints and target accuracy. These results demonstrate that our method optimally selects quick and accurate models that meet the target constraints and accuracy.

6.3 Validation of Runtime Detection with Adaptive Sequence

Lastly, we validate the dual-mode detection method to demonstrate that SecureRide achieves both high accuracy and responsiveness. Based on the scenario explained in Sections 5.1 and 4, we assume a driving period (from t_0 to t_T), with an illegal behavior occurring between t_s and t_f. t_T, t_s, and t_f are 100 s, 20 s, and 80 s, respectively. We divide the entire duration (from 0 to 100 s) into 0.2 s slots, resulting in a total of 500 slots. Then, we evaluate the accuracy and detection time by analyzing the decision of illegal behavior in each slot. For both speed violation and sidewalk detection, $\tau_{constraint}$ determining the quick model is set to 1, 1.5, and 2 s. α_{target} determining accurate model is set to 93, 94.5% for speed violation and 95, 99% for sidewalk detection. This results in nine cases. The number of hidden layers is limited to either 1 or 2.

Figure 9(a) and (b) shows the number of correctly predicted slots out of 500 for speed violation and sidewalk detection, respectively, across each case. The results indicate that for both behaviors, the effectiveness of our method is superior in Case #1–#3, while it diminishes in Case #–#9. The reason is that as the case number increases, the time constraints become alleviated, resulting in a smaller difference between Quick mode and Accurate mode. Blue boxes highlight our method's most effective cases, while red boxes show less effective ones.

Figure 9(c) illustrates the best case (Case 3) for sidewalk riding detection. In the figure, a value of 1 indicates that the illegal behavior is detected, while 0 indicates it is not. In this case, $\tau_{constraint}$ for determining the quick model is 1 s, and α_{target} for the accurate model is 99%. According to these settings, the quick model selected has the configuration HL 1, SL 5, and the accurate model has the configuration HL 2, SL 15. As shown in the figure, indicated by mark (1), the Quick mode generates considerable noise throughout the entire duration, not just within the illegal section, due to its lower accuracy (83.95%). Meanwhile, the Accurate mode detects the behavior stably with 99.59% accuracy but takes 3.09 s to identify the illegal behavior after it begins, as indicated by mark (3). The dual-mode detection achieves 1.01 s responsiveness (mark (2)) while maintaining 99.59% accuracy. This demonstrates that our method combines the low detection time of the quick model with the high accuracy of the accurate model, effectively leveraging the advantages of both modes. Furthermore, by raising α_{target}, SecureRide can employ a deeper accurate model (SL 25, HL 2) achieving up to 99.77% accuracy, while still benefiting from the low detection time (1.01 s) of the quick model under the dual-mode setting.

Figure 9(d) illustrates the worst case (Case 7) for sidewalk riding detection. In this case, $\tau_{constraint}$ is 2 s and α_{target} is 95%. According to these settings, the quick model is configured with HL 1, SL 10, and the accurate model is configured with HL 1, SL 25. Similar to the best case, Quick mode shows frequent misclassifications throughout the time window due to relatively low accuracy (88.12%). Notably, as indicated by mark (4), Dual mode exhibits fewer errors outside the illegal

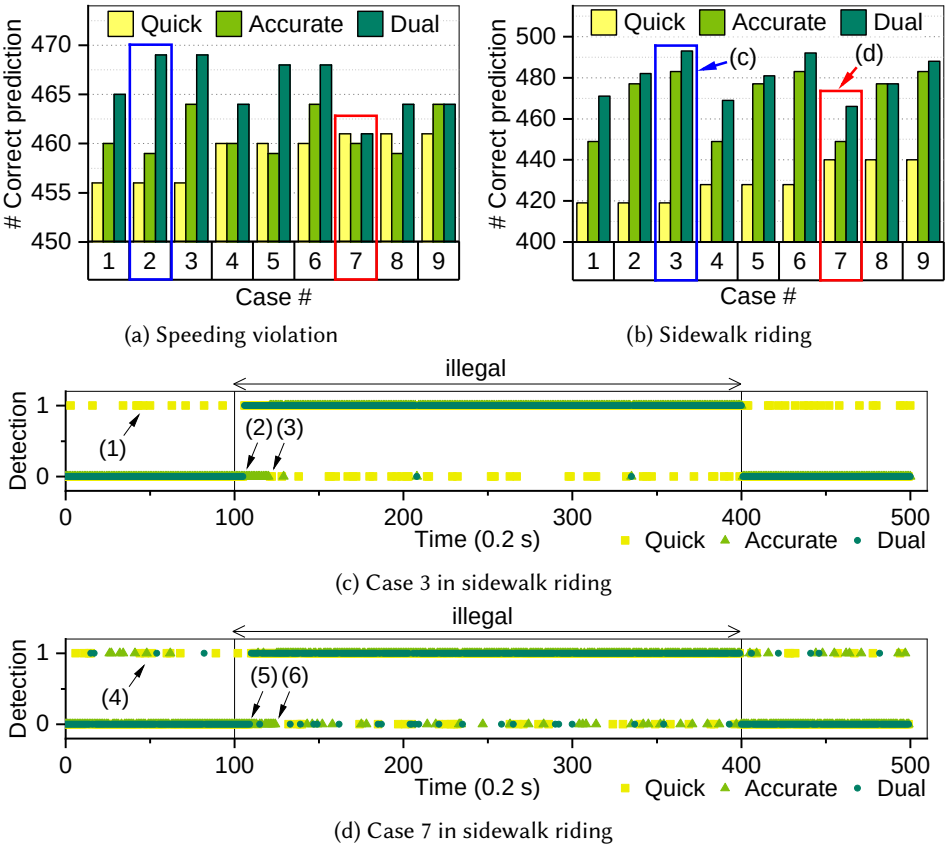


Fig. 9. Validation of dual-mode detection, which outperforms both quick and accurate modes.

region compared to Quick mode. This implies that the initial detection by Quick mode followed by confirmation via Accurate mode enhances reliability. As shown in mark (6), Accurate mode detects the illegal behavior stably with relatively high accuracy (95.12%) but with a detection time of 5.07 s. In contrast, as seen in mark (5), our dual-mode detection achieves a faster response time of 2.03 s while maintaining high accuracy. Although this case reflects a less pronounced benefit from the Dual mode due to a more relaxed $\tau_{constraint}$ and a lower α_{target} compared to other cases, it still demonstrates strong detection performance.

6.4 Comparison with GPS-based Solutions

To further evaluate the effectiveness of SecureRide, we conducted a comparative experiment against a GPS-based detection approach for sidewalk riding. Specifically, we aimed to assess how different urban environments affect GPS reliability by selecting three distinct driving scenarios. The first location, Park (811.5 m), is a road surrounding a park, where a sidewalk section begins halfway through. While the environment is generally open, there are occasional trees along the route. The second, Residential (659 m), has only the first approximately 150 m as paved road, followed by sidewalks. GPS accuracy in this area is affected by surrounding buildings. The third location, River (1,118.3 m), is a riverside bike path with no sidewalks along the entire route. As described in Section 6.1, ground-truth labels for sidewalk riding were manually recorded during the experiments. For the GPS-based method, we mapped the collected GPS traces to OpenStreetMap to identify

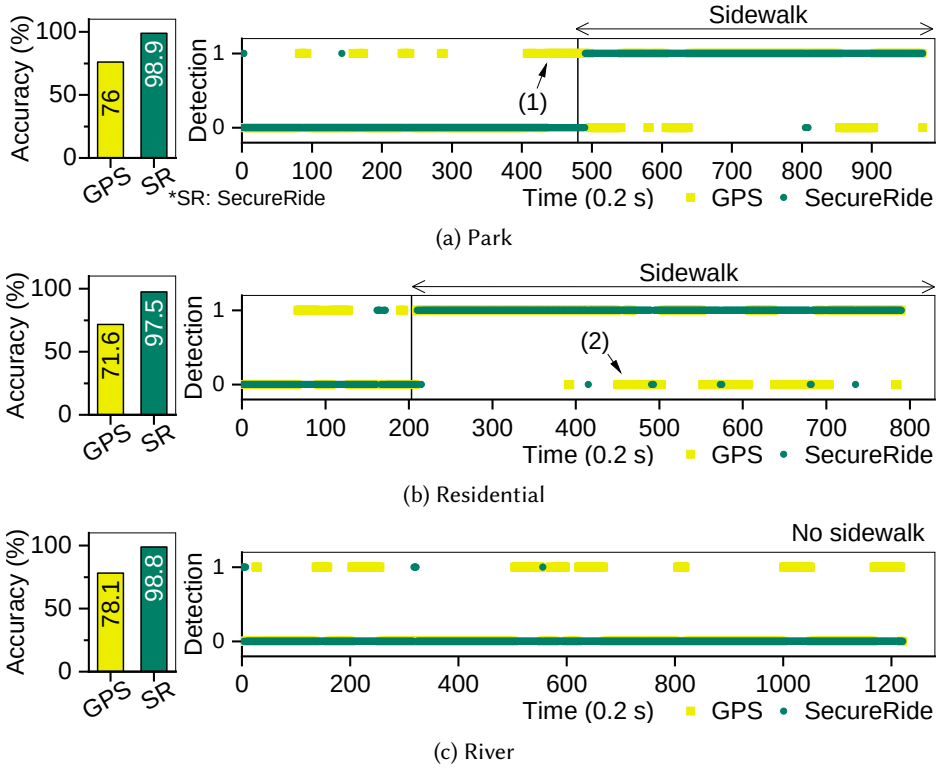


Fig. 10. Comparison with GPS-based detection for sidewalk riding. SecureRide significantly enhances detection accuracy compared to the GPS-based method.

whether the scooter was on a sidewalk. SecureRide detects sidewalk riding using battery information, which was collected at the time of driving. Note that we use the configuration from the best-case scenario in Section 6.3 (quick model: HL 1, SL 5; accurate model: HL 2, SL 15).

Figure 10 shows the results. In the figure, a value of 1 indicates that sidewalk riding is detected, while 0 indicates it is not. Figure 10(a) shows the results for the Park scenario. The GPS-based method exhibits frequent false positives and false negatives, particularly around the sidewalk transition zone (indicated by (1)), where proximity to sidewalks leads to unstable detection. In contrast, SecureRide consistently detects sidewalk riding with high reliability across the entire segment. Figure 10(b) presents the results in the Residential area. Due to the dense buildings and unstable GPS signals, the GPS-based method generates numerous errors. In particular, we observed a steep drop in accuracy when turning corners (indicated by (2)). SecureRide, on the other hand, maintained accurate detection throughout. Finally, Figure 10(c) illustrates the results for the River path, where no sidewalks exist. Despite the open environment, the GPS-based method frequently produced false detections, likely due to temporary signal instability. SecureRide correctly outputs no detections throughout the route, demonstrating a low false positive rate. Overall, these results show that SecureRide provides far more consistent and reliable detection than the GPS-based approach across diverse urban scenarios.

7 Related Work

E-scooter illegal behavior detection. There have been various studies and industry efforts to detect illegal e-scooter behaviors. Most approaches are based on utilizing GPS or additional sensors to solve

the problem. For example, Bird [5] implemented a method using high precision GPS modules to detect sidewalk riding. Similarly, Caggiani et al. [18] and Nikiforiadis et al. [20] proposed geofencing techniques that define restricted zones, enabling real-time monitoring of e-scooter usage on a larger scale in sharing systems. Although these methods are practical for real-time enforcement, GPS-based approaches-even with advanced sensors-suffer from accuracy degradation. Urban obstacles such as buildings and tunnels significantly interfere with GPS signals. Efforts using accelerometers, cameras, and vibration sensors have also been made. Voi [8] developed a solution employing deep learning models to differentiate between sidewalks using cameras on e-scooters. Kim et al. [17] and Ma et al. [19] attempt to monitor riding conditions based on vibration sensors. These approaches effectively detect illegal behaviors by sensing sudden accelerations, directional change, or proximity to pedestrians. However, they require additional sensor installation, which is impractical and cost-prohibitive for widespread use in commercial e-scooters. A recent study [26] has investigated the use of IMU sensors to extract various features from e-scooter riders' movements, applying contrastive learning. While effective at understanding rider actions, this method is not directly applicable to predicting illegal behaviors. In contrast to the above approaches, SecureRide provides an accurate illegal behavior detection system by utilizing battery factors, eliminating the need for additional sensors.

Behavior prediction using deep learning. Efforts have also been made to predict the behaviors of electric motor-driven vehicles, such as electric cars and drones, using deep learning. Since e-scooters also operate based on electric motors, these techniques have the potential to be applied to our problem. Firstly, for drones, Al-Haddad et al. [1], Jeon et al. [16], and Sadhu et al. [23] detect drone anomalies using CNN and LSTM models with IMU sensors, effectively capturing abnormal behaviors. However, these methods rely on high-performance IMU sensors required for precise PID control in drones, making them unsuitable for e-scooters with limited resources. Chen et al. [6] focused on diagnosing motor faults using only current signals, addressing classic deep learning challenges such as overfitting and accuracy loss in low-sample environments. However, this approach also requires heavy computation for use on e-scooter MCUs with limited resources. For electric cars, Alvarez et al. [9] predicts the energy efficiency of driving using multiple smartphone sensors. Sarker et al. [24] classifies the safety states of autonomous vehicles using speed, gas and brake pedal pressures, and steering wheel angles as inputs. These approaches rely on multiple sensors not available in e-scooters, limiting their applicability for e-scooter applications.

Novelty of SecureRide. SecureRide introduces two key innovations that distinguish it from prior work in both e-scooter behavior detection and embedded deep learning. First, SecureRide is the first system to detect illegal e-scooter behaviors using only internal battery information, eliminating the need for GPS, IMU, or other external sensors. By leveraging voltage, current, and capacity as behavioral indicators, SecureRide enables accurate detection while avoiding the cost, maintenance, and manipulation risks associated with traditional sensing methods. Second, we design a lightweight yet effective deep learning framework for behavior detection on resource-constrained e-scooter MCUs. Specifically, our design involves a lightweight model that uses sequences of multiple battery factors as input, and we formulate the problem as a binary classification. This structure preserves critical behavioral context while allowing for the deployment of lightweight neural networks on embedded MCUs. In addition, we devise a dual-mode detection strategy using heterogeneous DNNs. This approach resolves the tradeoff between detection accuracy and latency, achieving both timely and precise predictions.

8 Discussion and Limitation

We discuss potential challenges and practical considerations for real-world deployment and applicability of SecureRide.

8.1 Generalizability

While our system achieves high detection accuracy using a commonly used e-scooter in sharing systems, generalization to diverse e-scooter platforms and environments may require model adaptation. First, e-scooter products vary in wheel size and power consumption depending on motor characteristics, and their battery packs can operate within different voltage ranges. Therefore, retraining or fine-tuning the model with product-specific data may be necessary. In addition, battery aging and motor degradation over time can alter the voltage-current dynamics, potentially degrading model performance. Addressing this issue may require the integration of continuous learning techniques for long-term maintenance.

Second, although we evaluated our model across 13 distinct sidewalk paving patterns, there may be road surfaces where the visual or mechanical distinction between roadways and sidewalks is slight. In such cases, prediction accuracy may decline. A potential direction to mitigate this limitation is to incorporate collaborative sensing strategies, such as integrating GPS-based sidewalk maps or cross-verifying predictions among multiple e-scooters operating in the same vicinity.

Another important factor to consider is the distribution of training data. In our study, the dataset was collected under controlled conditions to ensure a balanced distribution of legal and illegal cases (see Tables 1 and 2). However, real-world deployments are likely to encounter significantly skewed distributions, where illegal behaviors are relatively rare. In such scenarios, the performance of the trained model may degrade. To address this, additional processing steps that consider such class imbalance, such as oversampling the minority class or undersampling the majority class, may be required during data preparation.

8.2 Applicability

SecureRide is designed for deployment in real-world mobility systems and provides multiple practical benefits that enhance its applicability. First, the detection results can be directly used by e-scooter operators to enforce safety rules in real time. When a violation is detected, the system can immediately alert the rider or automatically record the event for review. Based on these results, operators can take direct actions such as reducing the scooter's speed or notifying relevant authorities to respond to repeated offenses.

Second, SecureRide may support regulatory compliance by enabling automatic logging of detected violations along with timestamps and location data. These logs have the potential to serve as verifiable evidence for audits or legal processes, offering a way for municipalities and operators to monitor rule compliance without requiring continuous human oversight.

Finally, although SecureRide is evaluated on e-scooters, its underlying detection approach may be adaptable to other battery-powered mobility platforms. Devices such as electric bicycles (e-bikes), delivery robots, and other lightweight electric vehicles share similar battery characteristics and safety concerns. These similarities suggest the potential for applying SecureRide's methodology to a broader range of urban mobility systems, though further investigation and platform-specific adaptation would be required.

8.3 Scope of Illegal Behavior Detection Using Battery Information

We specifically target two behaviors—speeding and sidewalk riding—that are regulated in nearly all EU countries and U.S. states [10, 28]. Beyond these, a range of additional illegal behaviors have been defined across various countries and states. They include: riding with passengers (e.g., two persons on a single scooter), riding without a helmet (mandated in certain states or age groups), riding under the influence of alcohol, improper nighttime operation (e.g., no lights), and using a scooter under the minimum legal age. While the scope and strictness of these regulations vary by country and state, they collectively reflect growing global efforts to regulate e-scooter safety.

SecureRide relies on battery information to detect behaviors. As discussed in Sections 2.2 and 2.3, illegal behaviors that cause changes in power patterns may be detectable. For instance, riding with passengers can cause distinct power patterns due to increased load and changes in the center of gravity, which may lead to noticeable changes in battery voltage. Similarly, riding under the influence of alcohol could be detectable if intoxicated riders show unusual throttle or braking patterns that affect power usage. However, behaviors related to rider-specific attributes, such as helmet usage or rider age, do not impact the power delivery system and are therefore beyond the detection capabilities of SecureRide. While SecureRide has the potential to be extended to cover a broader range of illegal behaviors, it also has inherent limitations in detecting behaviors that are not reflected in the battery response.

8.4 Practical Deployment Considerations

We discuss key issues for practical deployment of our system. First, large-scale deployment is essential for practical use. SecureRide is designed with economic scalability in mind, as it relies solely on battery telemetry available through PMICs or BMS chips already embedded in most commercial e-scooters. This eliminates the need for additional sensors or hardware modifications, significantly reducing deployment and maintenance costs. The lightweight model variants can operate on MCUs such as the Arduino Nano, confirming compatibility with resource-constrained platforms commonly used in e-scooter fleets. These characteristics suggest that SecureRide can be integrated into existing systems with minimal overhead, making it well-suited for large-scale adoption.

Second, operational reliability and error handling are critical when detection results are used to support real-world interventions. As our system is designed to detect illegal behaviors, detection errors can lead to unintended consequences. For example, a false positive, in which legal behavior is mistakenly identified as illegal, may result in inappropriate enforcement if the detection result is used directly to impose penalties. Meanwhile, false negatives, i.e., missed detections, may allow dangerous behaviors to go unnoticed. If detection results are used to trigger safety interventions or enforce regulations, such misses could lead to delayed responses or a lack of accountability. While lightweight responses such as rider alerts may tolerate occasional misclassifications, stronger actions, including fines or usage restrictions, may require additional safeguards. To reduce this risk, the system can incorporate post-processing or secondary verification methods, such as confirming repeated violations over multiple frames, before taking decisive action.

9 Conclusion

E-scooters are an eco-friendly and convenient mode of transportation, but they pose a critical safety issue due to illegal behaviors. In this article, we propose SecureRide, a system that accurately detects illegal e-scooter behaviors at runtime using only battery information. To achieve this, we first design heterogeneous DNNs that predict the presence of illegal behaviors using three battery factors. We then propose a dual-mode technique for runtime operation, achieving both high accuracy and low detection time. We validated SecureRide through actual e-scooter driving experiments and tests on three embedded platforms. SecureRide operates without the need for additional sensors and is computationally efficient. We, therefore, expect that SecureRide can be seamlessly integrated into existing commercial e-scooter products and significantly contribute to mitigating global e-scooter safety concerns.

In future work, we aim to extend our detection framework beyond the two targeted behaviors—speeding and sidewalk riding—to address a broader range of unsafe or non-compliant riding patterns, such as tandem riding or aggressive acceleration. Additionally, our battery-based sensing approach holds potential for use in other contexts beyond illegal behavior detection, such as estimating

rider load, identifying mechanical anomalies, or analyzing usage patterns for fleet management. These extensions would further enhance the utility of SecureRide in both safety enforcement and operational intelligence.

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